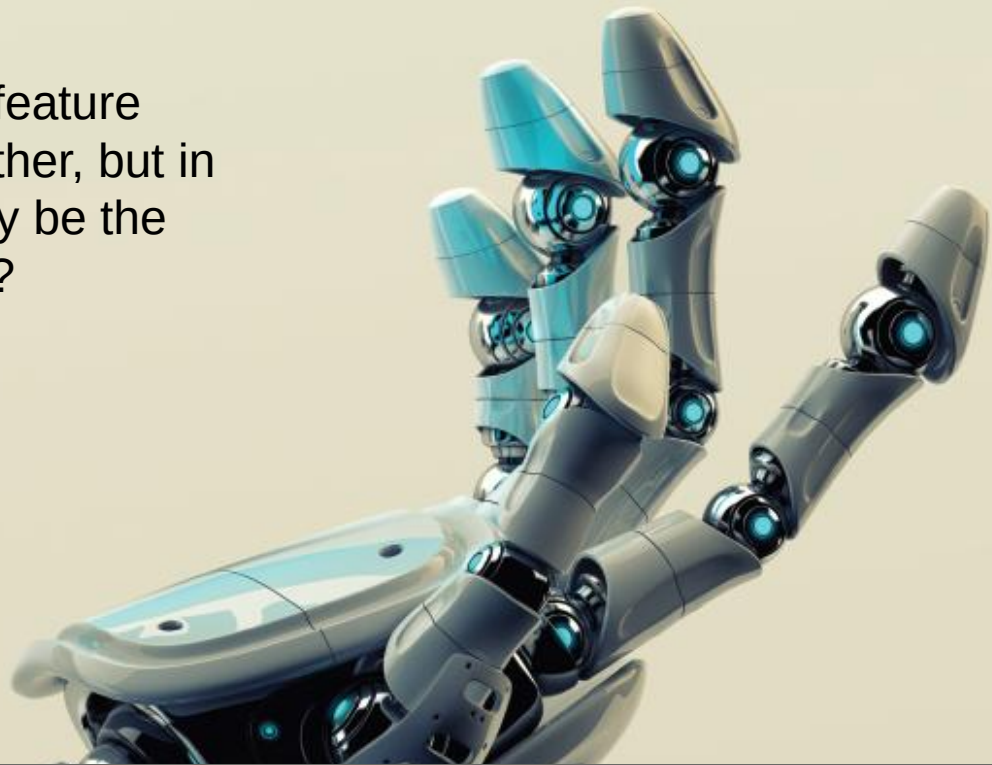




Unit 3

CYBORG TECH

Predictions of the future often feature robots and humans living together, but in reality, could cyborg technology be the next stage in human evolution?



GOALS

IN THIS UNIT, YOU WILL:

- Read about technologies that enhance human abilities.
- Learn about a technology that lets a person hear colors.
- Explore other devices that replace or improve human abilities.

THINK AND DISCUSS

1. What are some devices that help people see and hear better?
2. In what ways might technology help us experience the world in the future?

Lesson A

PRE-READING

- A.** Look at the photos on pages 42–45 and read the captions. What do you think each device does? Discuss your ideas with a partner.

Guesses will vary. Actual answers are:

Argus II:

a bionic eye that allows the wearer to see light by connecting the optic nerve to a special type of glasses

Cochlear implant:

a computer implant in the skull that connects to auditory nerves

EyeTap:

a device worn over the eyes that blocks out things the wearer doesn't want to see—such as an advertisement—and replaces them with something more interesting

B. Note answers to these questions and discuss your ideas with a partner.

1. Read the first paragraph on page 42. What is a cyborg? What are some examples of cyborgs?

A cyborg is a human whose abilities are improved by advanced technology.

2. Aside from the items pictured, what other cyborg technology do you think exists?

Answers will vary. Possibilities include: robotic body parts with mobility and superhuman strength, man-made transplant organs, artificial skin, etc.

Larry Hester testing out Argus II. The device includes a small video camera, a computer, a receiver, and an array of electrodes implanted on the patient's retina.



WE ARE CYBORGS

RoboCop, the Bionic Woman, Darth Vader—what do these characters have in common? They are all cyborgs—humans who are made more powerful by **advanced** technology. You might think that cyborgs exist only in fiction, or are a possibility only in the distant future. But cyborg technology already exists.



DEFINING CYBORGS

The word *cyborg*—a short form of “cybernetic organism”—was first used in a 1960 paper on space travel. A cyborg was defined as an organism “to which [external] **components** have been added for the purpose of **adapting** to new . . . environments.” According to this definition, an astronaut in a spacesuit is an example of a cyborg, as the spacesuit helps the astronaut adapt to a new environment—space. More recently, the word has evolved to refer to human beings who have **mechanical** body parts that make them more than human.



REAL-LIFE CYBORGS

Although super-humans like RoboCop are not yet a reality, advances in real-life cyborg technology allow some people to compensate for abilities they have lost, and give other people new and unusual abilities. An example is filmmaker Rob Spence and his bionic eye. Spence injured one of his eyes in an accident. Instead of replacing his damaged eye with the typical glass eye, Spence had a prosthetic eye with a camera **implanted** in his eye socket. The eye is not connected to Spence's brain or optic nerve, but it can record what Spence sees. Spence has used his camera eye to record interviews for a documentary about people with bionic body parts.

bionic: *adj.* using mechanical and/or electrical devices to increase a person's ability

cybernetic: *adj.* relating to the connections between biological, mechanical, and electronic systems

optic nerve: *n.* a nerve that sends visual information from the retina to the brain

prosthetic: *adj.* relating to an artificial body part



The x-ray (inset) shows the cochlear implant inside Michael Chorost's skull.



A more advanced type of bionic eye, called Argus II, was used for the first time by Larry Hester. A disease of the eye left Hester blind and in total darkness for over 30 years. But in September 2014, the 66-year-old was able to see again. The device consists of an implant in his left eye that connects his optic nerve to a pair of special glasses. When the device is turned on, it allows Hester to see light. Although his vision isn't completely restored, Hester can make out the shapes of objects, allowing him to see a world that had been in darkness for over three decades.

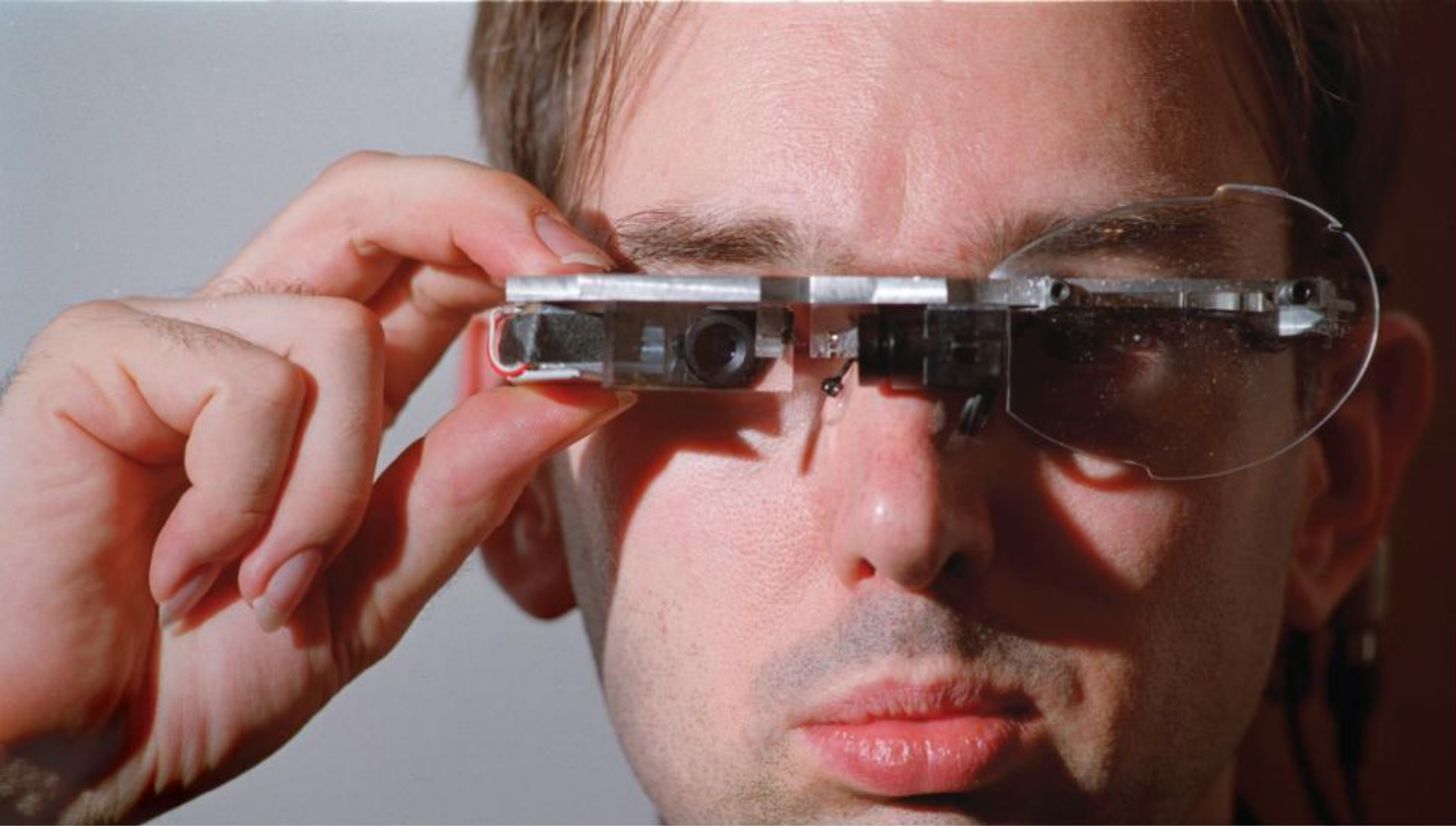
Some types of cyborg technology replace a lost ability by connecting directly to a person's nerves. Born partially deaf, Michael Chorost completely and suddenly lost his ability to hear in July of 2001. Two months later, doctors placed a cochlear implant, a kind of computer, inside his skull. This type of implant connects to auditory nerves and allows a deaf person to hear again. While Chorost's hearing isn't perfect, he continues to upgrade his implant as researchers improve the technology. Around the world, over 300,000 people have now been fitted with cochlear implants.



Steve Mann's cyborg technology changes an existing ability. In 1981, Mann, a researcher and inventor, created 36 kilograms (80 pounds) of wearable computer equipment that allowed him to change his reality. If Mann didn't want to see certain kinds of advertisements, his computer equipment would cover those advertisements with things that he actually wanted to see. So if he looked at a cigarette ad, for example, he would instead see an article that he wanted to read. In the 1990s, Mann reduced his computer equipment down to the size of a pair of glasses. Mann's device, the EyeTap, is now **permanently** attached to his head.

ARE WE ALL CYBORGS?

These examples of cyborg technology have enabled people to enhance or change their abilities and improve their lives. But does everyone want to **utilize** cyborg technology? It might be too late to decide. Cyborg anthropologist Amber Case argues that most of us are already cyborgs.



Steve Mann's cyborg technology—EyeTap—is permanently attached to his head.



Anyone who uses a computer or a smartphone, Case claims, is a cyborg. Consider the **data** that you have in your smartphone. It keeps information for you so you don't have to remember it: notes, phone numbers, email addresses, messages. It also allows you to communicate with friends and family via telephone, text messages, email, and social **networks** like Facebook, Twitter, Instagram, and Tumblr. Your computer and smartphone give you abilities that you would not otherwise have.

The potential benefits of cyborg technology are evident, but can this new technology be harmful, too? For example, when Steve Mann's EyeTap was forcibly **removed** at an airport, he had trouble walking and couldn't see correctly without it. The technology that had given him new abilities had apparently affected his abilities to walk and see. What might happen if, or when, more of us start to use devices like the EyeTap? Could we become too dependent on cyborg technology—and become less than human?

anthropologist: *n.* a scientist who studies people, society, and culture

auditory nerves *n.* nerves that send sound information from the ear to the brain



A THIRD EAR

Australian performance artist Stelios Arcadiou has taken cyborg technology to a completely different level. He has had a third ear (see page 46) surgically implanted on his arm. Why? As an artist, Arcadiou is interested in exploring what the human body can do. Unlike other prosthetic devices—which usually replace lost capabilities—Arcadiou’s implant adds something new to his body. The ear, which was reportedly grown in a lab, cannot currently hear. However, Arcadiou hopes to change that. He intends to have transmitters implanted in his third ear so that he will be able to hear through his arm!

Australian performance artist
Stelios Arcadiou shows off
his third ear.



Developing Reading Skills

GETTING THE MAIN IDEAS

- A.** Choose the best phrase to complete the first sentence of a summary of the passage.

Cyborg technology _____.

- a. will someday help people enhance their natural abilities
- ☒ b. is helping people enhance their natural abilities
- c. has a few benefits but also many serious drawbacks

B. Now complete the summary by checking (✓) the three sentences that best express the main ideas of the passage. The other sentences are either not main ideas or are not mentioned in the passage.

- ☒ a. Anthropologist Amber Case argues that computers and smartphones have turned most of us into cyborgs already.
- ☒ b. The term *cyborg* today refers to a person who has mechanical body parts that improve his or her performance.
- ☐ c. Born partially deaf, Chorost completely and suddenly lost his ability to hear in July of 2001.
- ☒ d. Cyborg technology is already helping people—by compensating for abilities they have lost and by enhancing existing abilities.
- ☐ e. Smartphones enhance our brains because they retain important information such as notes, phone numbers, email addresses, and messages.

UNDERSTANDING KEY DETAILS

A. Use the information from the passage to write short answers to these questions.

1. What device does Rob Spence's bionic eye contain? What does he use it for?

It has a camera in it that he uses to film what he sees. He uses it to record footage for a documentary.

2. How is the bionic eye used by Larry Hester different from Spence's?

Hester's bionic eye is actually connected to his optic nerve, whereas Spence's eye is not connected to his optic nerve or brain.

3. What human ability does Michael Chorost's device help him with?

The cochlear implant helps Chorost hear.

4. What does Steve Mann's device do?

It helps him block out images that he doesn't want to see, such as advertisements, and replaces them with interesting content, such as an article.

B. Use the information in the section “A Third Ear” to write a short answer to this question.

What does Stelios Arcadiou hope to do with his cyborg device?

He hopes to have transmitters implanted in the ear in his arm in order to be able to hear from it.

CRITICAL THINKING

1. **Analyzing.** Amber Case says that we are already cyborgs. What examples does she give to support her idea?

Case explains that the way we've introduced computers and smartphones into our daily lives has made us cyborgs, as both of these devices give us abilities that we otherwise wouldn't have.

2. **Evaluating.** Do you think that Case gives good examples of cyborg technology? Why or why not?

Answers will vary.

BUILDING VOCABULARY

A. Complete the information about bionics with the words in bold.

adapting implant mechanical network removed

Bionics is the study of 1. mechanical systems that function like living organisms or parts of living organisms. Bionic parts have helped thousands of people improve their lives. Here are some examples.

- Amanda Kitts lost her arm in a car accident. Doctors replaced it with a new, flesh-colored one. Underneath the plastic are three motors, a metal frame, and a(n) 2. network of sophisticated electronics. When Kitts thinks about moving her arm, sensors pick up impulses from her brain and turn motors to make the arm move.
- Aiden Keeney was born deaf. As a baby, he got cochlear implants in both ears. He is 3. adapting well to his bionic devices. As a result, he has a good chance of developing clear speech, his doctor says.
- Researchers are planning to develop an artificial hippocampus, the part of the brain that stores memories. They will 4. implant it in the brains of people with memory loss.

While astonishing, it's worth remembering that not all of these devices work perfectly. Some patients have had their bionic devices 5. removed because they interfere with other medical devices that the patients depend on, such as pacemakers (devices that regulate the heartbeat).



B. Match the bold words with their definitions.

- | | |
|---|---------------------------------|
| 1. <u> e </u> Being able to hear changed Chorost's life permanently . His life has been different ever since he had a cochlear implant. | a. a part of something |
| 2. <u> d </u> Amanda Kitts is able to utilize her bionic arm in many of the same ways that she did with her natural arm. | b. raw or organized information |
| 3. <u> b </u> Modern smartphones hold an incredible amount of data . If we lost all our contacts and notes, we might feel very confused. | c. complex, difficult |
| 4. <u> a </u> A hearing aid is an example of an external component that helps a person hear. | d. to use |
| 5. <u> c </u> Someday, people may have bionic devices that help them move objects with their minds. This is an example of very advanced cyborg technology. | e. forever |

GETTING MEANING FROM CONTEXT

Learning how to use the context—the surrounding words—to guess the meanings of unknown words helps increase your reading speed. It's also useful in situations where you do not have a dictionary, such as when you're taking a test.

Find the following sentence in the section “A Third Ear” on page 45. Then answer the questions below.

“Australian performance artist Stelios Arcadiou **has taken** cyborg technology **to a completely different level.**”

1. How is Stelios Arcadiou's ear different from the other devices you read about? Would you describe him as a “cyborg”? Why or why not?

Arcadiou does not need the ear to help him with a deficiency; instead, he simply wants to add to what his body already has. Perhaps once the ear has the ability to hear, he can be called a cyborg.

2. What do you think the expression “to take to a new level” means?
 - a. To put something in a completely different place
 - ☒ b. To do something in a different or more advanced way

CRITICAL THINKING

Evaluating. Look again at the last paragraph of the passage. Could we become too dependent on cyborg technology, in your opinion? What are some possible drawbacks of cyborg technology? Discuss your ideas with a partner.

Answers will vary.

EXPLORE MORE

Read more about Amber Case's ideas at nationalgeographic.com/explorers. What does Case compare smartphones to? How does she describe the way people feel when they lose their smartphones? Share your information with the class.

I LISTEN TO COLOR



NEIL HARBISSEON Artist and Cyborg Activist, TED speaker

Neil Harbisson was born without the ability to see colors. He explains, “To me, the sky is always gray, flowers are always gray, and television is still in black and white.”

As a young adult, Harbisson studied music and art, and all his early artistic work was in black and white. Then a lecture on cybernetics gave him the idea for a device that would allow him to finally experience a world of color. The device, called the Eyeborg, matches colors with sounds. As a result, Harbisson is now able to create art using color.

Harbisson is now a cyborg activist. He created the Cyborg Foundation to help people who want to become cyborgs, to defend the rights of people with cyborg devices, and to promote cyborg research.

activist: *n.* a person who works at changing something, especially in politics

In this lesson, you are going to watch segments of Harbisson's talk. Use the information above to answer each question.

1. Why was all of Harbisson's early artwork in black and white?

Harbisson is color-blind. Since he can't see colors, his artwork lacked colors, too.

2. What does the Eyeborg match colors with? What did it enable Harbisson to do?

It matches colors with sound, which allows Harbisson to hear a different sound for each color.

3. What do you think a "cyborg activist" is? In addition to art, what are some of Harbisson's interests?

As a cyborg activist, Harbisson helps people who want to become cyborgs, defends the rights of people with cyborg devices, and promotes cyborg research. He also educates the general public on the benefits of cyborg technology. In addition to art, Harbisson is clearly also interested in being active with issues related to cyborg technology.

Neil Harbisson's **idea worth spreading** is that when we enhance our senses with cyborg technology, we can extend our knowledge of the world around us.

PREVIEWING

- A. Read this excerpt from Neil Harbisson's talk. What verbs do you think are missing? *Guesses will vary. Actual answers are:*

"Well, I was born with a rare visual condition called achromatopsia, which is total color blindness, so I've never seen color, and I don't know what color 1. looks like, because I come from a grayscale world. To me, the sky is always gray, flowers are always gray, and television is still in black and white.

But, since the age of 21, instead of 2. seeing color, I can hear color. In 2003, I started a project with computer scientist Adam Montandon, and the result . . . is this electronic eye. It's a color sensor that detects the color frequency in front of me—and 3. sends this frequency to a chip installed at the back of my head, and I 4. hear the color in front of me through the bone, through bone conduction."

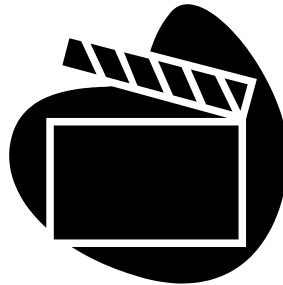
color frequency: *n.* the number of light waves that pass a point in space in one second. This number is different for each color.

(bone) conduction: *n.* moving through (bone)

- B.** What do you think a “grayscale world” is? Discuss your ideas with a partner.

Answers will vary. Harbisson is explaining that when he looks around, he only sees white, black, and the various shades of gray in between.

- C.** Check your ideas above as you watch (▶) the first segment of the TED Talk.



UNDERSTANDING KEY DETAILS

Complete the following sentences about the information in this part of the talk.

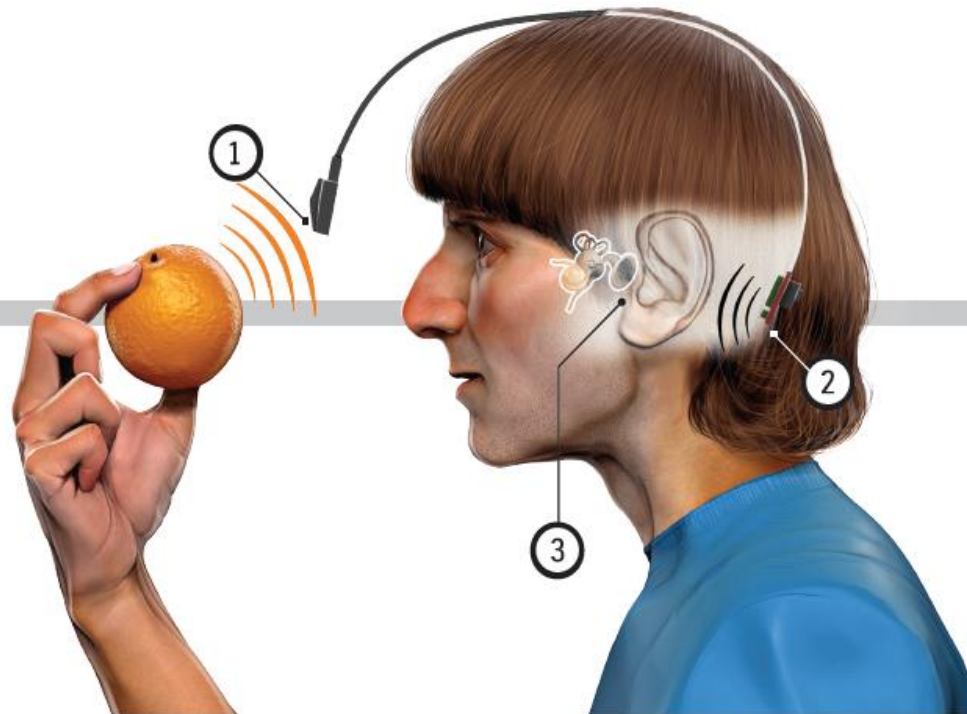
1. Harbisson's device helps him to hear color.
2. When he first started using his device, Harbisson needed to remember the names of colors and their notes.
3. When he started dreaming in color, Harbisson felt that the software and his brain had finally united and he started to feel like he was a cyborg.

UNDERSTANDING A PROCESS

THE EYEBORG

Understand how the device implanted in Harbisson's head transforms color into sound.

① A sensor detects the frequency of the color in front of Harbisson and transmits it through a chip installed on the back of his head.



② The chip converts the colors into sound waves. Each color corresponds to a musical note.

③ These sound waves travel through the skull using bone conduction, and arrive at Harbisson's auditory system.

Study the infographic that shows how the Eyeborg works. Then complete the sentences.

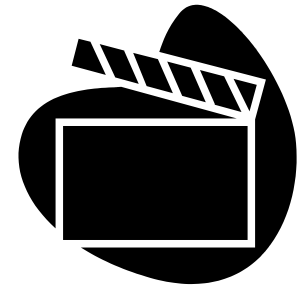
Harbisson wears a sensor that sits in front of his head. The sensor connects to a chip that is 1. installed in the back of Harbisson's head. The chip converts color frequencies to 2. sound waves. The sounds that the chip produces 3. travel to his auditory nerves.

PART 2

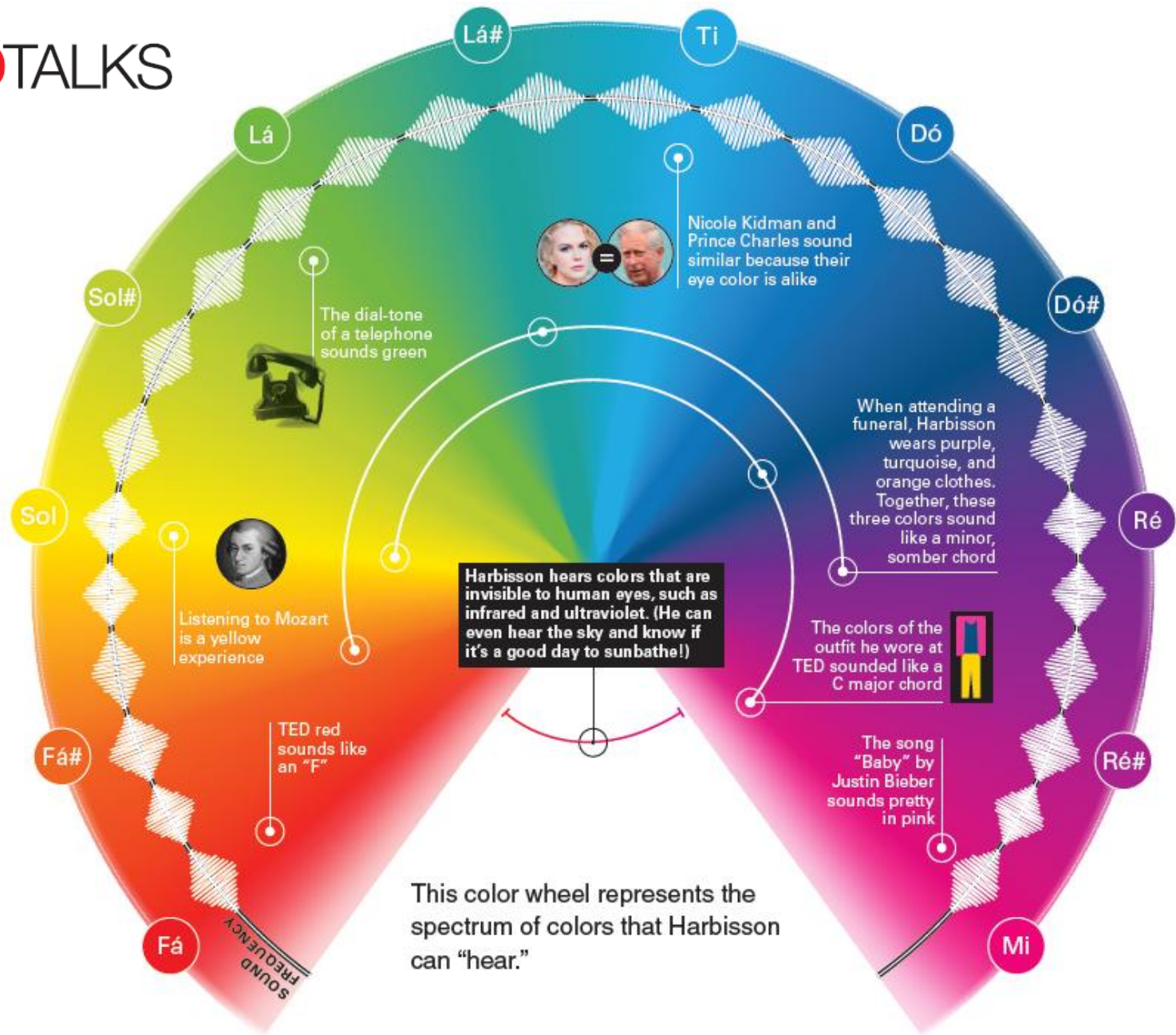
A LIFE-CHANGING DEVICE

UNDERSTANDING KEY DETAILS

Watch (▶) the next segment of Harbisson's TED Talk. Use information from this part of the talk and the infographic on page 54 to complete the summary below.



Harbisson enjoys creating 1. pictures of people. Instead of drawing someone's face, he points at them with the Eyeborg and writes down the different 2. notes he hears. Eventually, he was able to distinguish 360 3. colors, just like human vision. But there were many more colors that the human eye could not detect. He decided to extend his color senses, and added 4. infrared and 5. ultraviolet to the color-to-sound scale, so now he can hear colors that the human eye cannot perceive.



CRITICAL THINKING

1. **Reflecting.** How do you think the Eyeborg improves Harbisson's life? What are some possible drawbacks to the Eyeborg?

Answers will vary. Harbisson's world has become brightened by the sound of all the colors around him. He seems to genuinely enjoy using the Eyeborg. Some drawbacks include possible limitations of the device, for example, perhaps he can't go in water easily. Others might say he is too dependent on the sounds of the device and that it takes away from his natural experiences.

2. **Discussion.** Think back to the types of cyborg devices that you read about in Lesson A. Which cyborg device is the most similar to Harbisson's? In what ways is it similar? Discuss your answer with a partner.

Answers will vary. Each device is similar to the Eyeborg in certain ways. Spence's bionic eye influences his work and art in a similar way that Harbisson's does. The Argus II provides an alternative for a lost ability as the Eyeborg does for Harbisson. In a similar way, the cochlear implant helps replace a lost ability. And the Eyeborg also changes an existing ability in the same way Mann's EyeTap does.

EXPLORE MORE

Find out more about Harbisson's device. Watch the rest of his talk at TED.com. What are some other ways his device affected his life and his work? Share your information with the class.

Project



Tan Le demonstrates a new computer interface that can read its user's brainwaves, making it possible to control virtual objects with mere thoughts.

Project

A. Work with a partner. You are going to give a presentation on another example of cyborg technology.

1. Go to TED.com. Watch some of the following TED Talks to get ideas:

- Tan Le, “A headset that reads your brainwaves”
- Pranav Mistry, “The thrilling potential of SixthSense technology”
- Hugh Herr, “The new bionics that let us run, climb and dance”
- Eythor Bender, “Human exoskeletons—for war and healing”
- Aimee Mullins, “My 12 pairs of legs”
- Todd Kuiken, “A prosthetic arm that ‘feels’”

Project

2. With your partner, choose one of the devices in the TED Talks you saw or think of your own idea for a new cyborg technology. Answer these questions about it.
 - What is it made of? How does it work?
 - How is it connected to the body or the brain?
 - Does it replace or enhance an existing ability, or does it give a completely new ability?
 - How does it improve the user's quality of life?
3. Use your information to create a two-minute presentation. You can use drawings, photos, and video to explain your information.

B. Work with two other pairs.

- Give your presentations.
- As you listen, take notes.
- At the end, review your notes.
- Discuss: Which device is the most interesting? The most useful? Why?

Project

EXPLORE MORE

Learn more about cyborg technology at nationalgeographic.com. Find out how scientists are using cyborg technology with animals and insects. Share your information with the class.

Part 1

Well, I was born with a rare visual condition called achromatopsia, which is total color blindness, so I've never seen color, and I don't know what color looks like, because I come from a grayscale world. To me, the sky is always gray, flowers are always gray, and television is still in black and white.

But, since the age of 21, instead of seeing color, I can hear color. In 2003, I started a project with computer scientist Adam Montandon, and the result, with further collaborations with Peter Kese from Slovenia and Matias Lizana from Barcelona, is this electronic eye. It's a color sensor that detects the color frequency in front of me—and sends this frequency to a chip installed at the back of my head, and I hear the color in front of me through the bone, through bone conduction. So, for example, if I have, like—this is the sound of purple. For example, this is the sound of grass. This is red, like TED. This is the sound of a dirty sock. Which is like yellow, this one.

So I've been hearing color all the time for eight years, since 2004, so I find it completely normal now to hear color all the time. At the start, though, I had to memorize the names you give for each color, so I had to memorize the notes, but after some time, all this information became a perception. I didn't have to think about the notes. And after some time, this perception became a feeling. I started to have favorite colors, and I started to dream in colors.

So, when I started to dream in color is when I felt that the software and my brain had united, because in my dreams, it was my brain creating electronic sounds. It wasn't the software, so that's when I started to feel like a cyborg. It's when I started to feel that the cybernetic device was no longer a device. It had become a part of my body, an extension of my senses, and after some time, it even became a part of my official image.

This is my passport from 2004. You're not allowed to appear on U.K. passports with electronic equipment, but I insisted to the passport office that what they were seeing was actually a new part of my body, an extension of my brain, and they finally accepted me to appear with the passport photo. . . .

Part 2

So I really enjoy creating, like, sound portraits of people. Instead of drawing someone's face, like drawing the shape, I point at them with the eye and I write down the different notes I hear, and then I create sound portraits. Here's some faces.

Yeah, Nicole Kidman sounds good. Some people, I would never relate, but they sound similar. Prince Charles has some similarities with Nicole Kidman. They have similar sound of eyes. So you relate people that you wouldn't relate, and you can actually also create concerts by looking at the audience faces. So I connect the eye, and then I play the audience's faces. The good thing about this is, if the concert doesn't sound good, it's their fault. It's not my fault, because . . .

And so another thing that happens is that I started having this secondary effect that normal sounds started to become color. I heard a telephone tone, and it felt green because it sounded just like the color green. The BBC beeps, they sound turquoise, and listening to Mozart became a yellow experience, so I started to paint music and paint people's voices, because people's voices have frequencies that I relate to color.

And here's some music translated into color. For example, Mozart, "Queen of the Night," looks like this. Very yellow and very colorful, because there's many different frequencies. And this is a completely different song. It's Justin Bieber's "Baby." It is very pink and very yellow. . . .

So I got to a point when I was able to perceive 360 colors, just like human vision. I was able to differentiate all the degrees of the color wheel. But then, I just thought that this human vision wasn't good enough. There's many, many more colors around us that we cannot perceive, but that electronic eyes can perceive. So I decided to continue extending my color senses, and I added infrared and I added ultraviolet to the color-to-sound scale, so now I can hear colors that the human eye cannot perceive. . . .

We should all think that knowledge comes from our senses, so if we extend our senses, we will consequently extend our knowledge. I think life will be much more exciting when we stop creating applications for mobile phones and we start creating applications for our own body. I think this will be a big, big change that we'll see during this century.

So I do encourage you all to think about which senses you'd like to extend. I would encourage you to become a cyborg. You won't be alone. Thank you.